

Taha Bekar

Robotics Software Engineer — Perception — SLAM — Autonomous Systems
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Education

Rensselaer Polytechnic Institute M.S. Electrical, Computer, and Systems Engineering Focus: Autonomy, Robotics, and Control	Troy, NY 2024 – May 2026
Middle East Technical University B.S. Electrical and Electronics Engineering GPA: 3.56 / 4.00	Ankara, Turkey 2018 – 2024

Research Experience

Deformable Object Manipulation Research – RPI Research Assistant / Master's Project	2025 – Present
- Developed a ROS2-compatible deformable-object simulation framework in NVIDIA Isaac Sim using Newton Style3D for real-time cloth manipulation with anisotropic material, contact, friction, damping, and bending parameters. - Built an obstacle-aware cloth manipulation pipeline with kinematic corner pinning, simulated cloth state publishing, ROS2 command interfaces, obstacle-distance computation, and closed-loop follower-corner control. - Integrated a Control Barrier Function safety controller for cloth-obstacle avoidance for contact-rich manipulation scenarios. - Developed real-time perception and state estimation pipeline for planar deformable objects using RGB-D camera data, ArUco marker tracking, and Kalman filtering in ROS for real-to-sim calibration.	

Work Experience

Kuartis Computer Vision / Machine Learning Engineer	Ankara, Turkey Feb 2024 – Jun 2024
- Evaluated LiDAR point cloud registration algorithms including ICP, NDT, GICP, and FAST-GICP for SLAM pipelines and real-time mapping systems. - Implemented LiDAR-IMU sensor fusion by combining LiDAR odometry with IMU odometry, evaluating both tightly coupled and loosely coupled fusion strategies for improved localization robustness. - Analyzed performance of SLAM frameworks including LOAM, LIO-SAM, and FAST-LIO for localization accuracy and computational efficiency. - Improved LIO-SAM localization accuracy by 6% by integrating GICP-based scan matching into the GTSAM factor-graph optimization framework.	
GT-ARC Gemeinnützige GmbH, DAI Labor, TU Berlin Software Engineer Intern	Berlin, Germany Jul 2023 – Sep 2023
- Developed LiDAR-based free parking space detection algorithm for autonomous vehicles using ROS, Autoware, and HD Maps by implementing geometric intersection analysis between detected vehicles and parking regions. - Achieved 86% free space detection accuracy.	
Analog Devices Software Engineer Intern	Istanbul, Turkey Jul 2022 – Aug 2022
- Implemented PWM controller firmware on MAX32666 microcontroller using timer peripherals. - Built GUI interface for testing PWM signals and implemented unit tests using Google Test.	

Projects

Autonomous UGV Mapping System
- Converted a 2D LiDAR into a 3D scanning system via servo rotation to enable volumetric mapping. - Implemented octomap mapping using ICP (Iterative Closest Point) registration and AMCL (Adaptive Monte Carlo Localization) algorithms in ROS. Achieved accuracy levels of 95% and 75% for mapping and localization in ROS.
Lane Detection System
Implemented classical vision-based lane detection using Canny and Hough transform.
FPGA Network QoS Controller
Implemented priority-based packet scheduling system in Verilog.

Technical Skills

Programming: Python, C++, C, MATLAB
Robotics: ROS, ROS 2, SLAM, NVIDIA Isaac Sim, MuJoCo, PhysX, Newton, Gazebo, Motion Planning, Localization, Kinematics
Perception: Computer Vision, LiDAR Point Cloud Processing
Libraries: PCL, OpenCV, PyTorch, NumPy
Tools: Linux, Git, Docker

Relevant Coursework

Safe Autonomy Computational Vision Machine Learning Robotics II Computational Optimization